# Say Swear

**Stage 2 Implementation Report**

EECS 3311 - Fall 2026 | Status date: November 7, 2026 (Toronto)

**Review draft - submission evidence is still pending.** This report describes the public Stage 2 implementation. The user reports having played the game and confirms that it works properly. The instructor confirmed that the existing early commit history works for weekly-progress evidence. The feature video and timestamps are supplied, with playable YouTube metadata and matching duration confirmed November 7. Focused source review remains pending; AI usage and cost figures are provisional student estimates where stated. Submission is due November 10, 2026.

This report accounts for all ten Stage 1 features, explains the five implemented patterns, and connects the current Java source to the revised UMLet diagrams. It separates recorded verification from evidence that still needs to be supplied.

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# 1. Project overview and implementation status

## Problem, users, and scope

Say Swear is a small 2D precision game for players who want to navigate a narrow winding path using natural-language instructions. One player character moves over a void toward one goal. Leaving the safe path cancels movement and returns the character to the level start. The implemented scope contains one fixed level, four-direction movement, combined directions, bounded timed movements, falling, and start recovery. There are no checkpoints, enemies, jumping, inventory, procedural generation, autonomous navigation, or pathfinding.

The human chooses every turn and its timing. A JavaFX GUI displays the actual level and accepts continuous microphone input, with a text-instruction field for the same interpreter. A Java terminal CLI accepts natural-language text and displays a map, coordinates, velocity, held directions, transcript/action feedback, and lifecycle. Both frontends use the same Java 21 controller, game, agent, and control objects composed by `Main.compose()`; the CLI substitutes typed text for capture/transcription.

## AI integration and deterministic execution

GUI audio goes through `VoiceInputService` and `StreamingSpeechRecognizer`/`SherpaOnnxAdapter`. sherpa-onnx produces changing partial transcripts and endpoints. Audio is not streamed to SemIf. Eligible text and a frozen `ControlContext` go through `ApplicationController`, `VoiceCommandAgent`, and `DecisionModel`/`OpenJevAdapter` to the local SemIf service running Qwen3.5 4B. Python hosts that external inference runtime; the application, simulation, and frontends are Java.

SemIf scores finite alternatives from real model logits. The bridge composes one of 52 bounded semantic labels across eight action types, including `REPEAT_LAST`. The model receives transcript text, current held directions, prior accepted text, structured last movement, and movement eligibility. Java retains epoch/version/revision identity locally and stamps it on the returned decision. The request contains no map, position, route, start, or goal coordinates. The model interprets the human's command and selects a bounded intent. It does not directly call the game or choose how to reach the goal.

`ControlManager` checks schema, session epoch, transcript version, context revision, and player eligibility. It plans deterministic commands, cancels obsolete work, and executes through `ActionExecutor` and `GameCommand`. Java owns timer duration/expiry, repeat-memory resolution, conflict validation, simulation, collisions, observer cleanup, and fall recovery. `MovementIntent` remains available after a timed action releases its input, so a fresh `again` repeats the remembered movement rather than the release action.

The shared application loop serializes game/control changes. Worker threads handle microphone and HTTP work; JavaFX receives coalesced presentation updates. The current decision deadline is 2000 ms including agent queue time. New instructions supersede older requests; stale successes and failures cannot affect a newer session. The explicit `--demo` option uses a labelled deterministic parser for wiring checks. It is not an LLM and is never an automatic fallback.

## Implementation and evidence status

| Area | Current status | Evidence or remaining work |
|---|---|---|
| F01-F10 and both frontends | Implemented, with documented refinements | Complete feature/method/pattern mapping in section 2; genuine model and deterministic paths are distinguished. |
| Five course patterns | Implemented | Section 3 explains concrete participants and source methods; Appendix A contains the updated class diagram. |
| User gameplay | Acceptance and recorded demonstration | The November 6 recording shows GUI voice/text gameplay and CLI play, with feature timestamps in section 7. Detailed source/test review is not implied. |
| Ordinary Java verification | Recorded pass | November 7: 92 tests, zero failures/errors/skips; build/installDist passed after five separate defect fixes. |
| Live and platform coverage | Partial recorded evidence | Native sample-audio ASR, historical real-model checks, and packaged CLI checks exist. GUI microphone capture and partial transcripts are recorded. Detailed device-failure/recovery, native Linux, and updated repeat-enabled 2000 ms GPU performance remain unverified. |
| AI tool accounting | Provisional student report | Approximately 6 billion Codex tokens and CAD $282.50 subscription spending are reported; local inference tokens and compute costs are unmeasured. Confirm period, currency, and allocation. |
| Weekly progress evidence | Five fixes pushed November 7; early history accepted | Section 6 links five meaningful code/test commits for period 3 and records the instructor's acceptance of earlier development for periods 1-2. Commit dates are unchanged. |
| Submission evidence | Partly pending | Video links/playable metadata are checked. Source-review outcomes, AI accounting and the SD04 timing-order refinement remain to complete. |

## Reproduction and configuration

The repository includes a Gradle wrapper, Java dependencies, GUI/CLI launchers for Windows and Linux, and pinned setup instructions for SemIf and streaming ASR. Follow the root README, SemIf setup, and ASR setup. Start the real local service, wait for readiness, then launch GUI or CLI without demo mode. CLI play needs neither a microphone nor an ASR model.

Local inference requires no API key. `.env.example` contains safe environment-variable examples; the application does not auto-load `.env`. Configuration can be supplied through environment variables or launcher options. `.gitignore` excludes local environment files, downloaded weights, caches, logs, and build outputs. Report aggregate usage/cost totals with their accounting basis; private billing documents, account identifiers, and machine-specific paths are not needed.

Source references below are repository-relative paths. The supplied UMLet SVGs are preserved unchanged; their large-format PDF appendix pages allow readable zooming. Section 7 provides timestamped YouTube links; playable metadata and matching duration were checked November 7.

# 2. Feature to implementation mapping

All ten original feature IDs are accounted for. F10 replaces checkpoint recovery with return-to-start recovery. Repeat memory extends F05/F08/F09. Supporting input/session/display refinements are documented as changes, not counted as additional features.

### Maintained feature mapping

This mapping follows the Course Project Stage 2 Instructions: every Stage 1 feature is accounted for with its status, implementation files/responsibilities, key methods, patterns, behavior type and video reference. The revised Stage 1 report retains F01-F10 and UC01-UC06. The original design is preserved in Git history at commit `56da5fd95c43dde4760032e59a36787a2c74049c`; current UMLet SVG diagrams describe the implementation. Design changes explain differences.

All ten features have connected implementations. The November 6 recording and feature timestamps show GUI microphone input, streaming text, typed interpretation, contextual replacement, combined/timed actions, repeat, cleanup, fall recovery and CLI gameplay. On November 7, YouTube returned playable status and a 691-second duration matching the reviewed recording. Demonstrations are finite samples; detailed device-failure coverage, native Linux operation and the updated 2000 ms GPU gate remain unverified.

## Feature index

| Feature | Stage 1 reference | Status/change | Type | Video |
|---|---|---|---|---|
| F01 - Real-Time Precision Movement | F01, same requirement | Implemented; CLI presentation refined | Deterministic | 01:05-03:25 |
| F02 - Continuous Voice-Control Session | F02, same requirement | Implemented; lifetime/status refined | Deterministic session management | 01:23-03:25 |
| F03 - Streaming Speech Transcription | F03, same requirement | Implemented | AI-based; Java delivers text | 01:44-02:08 |
| F04 - Natural-Language Movement Interpretation | F04, same requirement | Implemented; local SemIf/Qwen provider | Hybrid | 03:44-04:30 |
| F05 - Context-Aware Commands | F05, same requirement | Implemented; repeat memory added | Hybrid | 04:35-05:37 |
| F06 - Live Command Correction / Interruption | F06, same requirement | Implemented; bounded scheduling | Hybrid | 04:35-05:37 |
| F07 - Combined Directional Commands | F07, same requirement | Implemented | Hybrid | 05:37-06:23 |
| F08 - Magnitude / Timed Commands | F08, same requirement | Implemented; timed replay added | Hybrid | 06:23-07:41 |
| F09 - Control-State and Conflict Management | F09, same requirement | Implemented; freshness/replay guards | Deterministic | 07:47-08:17 |
| F10 - Fall Detection and Start Recovery | F10 - Fall Detection and Checkpoint Recovery | Modified: every fall returns to start | Deterministic | 08:18-09:20 |

GUI text entry, explicit demo mode, session controls, repeat memory, terminal redraw and Linux setup are accounted for in D04-D07 and D18-D20. They extend these features or support setup and are not counted as additional features to reach ten. Checkpoint progression is explicitly removed under D17; no original feature ID is silently omitted.

## F01 - Real-Time Precision Movement

**Reference/status:** Stage 1 F01; implemented, with D06/D10/D19 presentation/timestep refinements. **Design:** UC01/UC02; SD02/SD05. **Patterns:** MVC, Observer, State. **Type:** deterministic.

**Files, responsibilities and key methods:** ApplicationController schedules `tickFromClock()`/`advance()`. GameModel owns `update()`/`evaluatePosition()`. ActiveState subdivides `update()` for collision checks; Player applies `move()` and normalized velocity. Level supplies `isSafe()`/`isGoal()`. JavaFxGameView and CliView implement snapshot `render()`.

**Input -> collaboration -> result:** accepted directional controls set the game's held inputs. Loop ticks advance the player, check the corridor and publish snapshots. JavaFX draws the level/player; the CLI presents an adaptive map and exact position/velocity, with supported in-place dashboards refreshed at up to 10 Hz. Neither view simulates movement. Diagonal speed is normalized.

**Verification:** saved movement/geometry/goal tests and terminal tests pass; historical GUI/packaged CLI checks exercised actual movement. **Video:** 01:05-03:25.

## F02 - Continuous Voice-Control Session

**Reference/status:** Stage 1 F02; implemented, refined by D11/D12. **Design:** UC01; SD01. **Patterns:** MVC, Adapter. **Type:** deterministic session management around ASR.

**Files, responsibilities and key methods:** JavaFxGameView routes `requestStartVoice()`/`requestStopVoice()`. ApplicationController coordinates `startVoiceControl()`, `stopVoiceControl()`, `handleVoiceReady()` and `handleVoiceFailure()`. VoiceInputService owns `start(epoch)`, capture-worker `run(session)`, `stop()` and `close()`. MicrophoneDevice opens Java Sound and supplies `readSamples()`, which the service's capture worker calls repeatedly.

**Input -> collaboration -> result:** Start listening creates a session and initializes recognition on a worker. After the microphone opens, readiness changes the visible status. Capture continues across utterances until stop/failure. Stop & release invalidates old work and releases controls/capture. CLI text substitutes for this audio entry point.

**Verification:** six controlled voice-service tests cover readiness, epoch isolation, nonblocking startup/stop, failure and PCM conversion. The recording shows a real GUI listening session; detailed device-failure and repeated recovery coverage remain unverified. **Video:** 01:23-03:25.

## F03 - Streaming Speech Transcription

**Reference/status:** Stage 1 F03; implemented. **Design:** UC01; SD01. **Patterns:** Adapter, MVC. **Type:** AI-based recognition with deterministic delivery.

**Files, responsibilities and key methods:** StreamingSpeechRecognizer defines `start()`/`acceptAudio()`/`stop()`. SherpaOnnxAdapter loads the online model and decodes incrementally in `acceptAudio()`. VoiceInputService forwards `partial()` and advances utterance identity in `endpoint()`. TranscriptUpdate carries identity; ApplicationController.handleTranscript() versions submissions. JavaFxGameView.showTranscript() displays changing words.

**Input -> collaboration -> result:** successive audio chunks produce changed partial text before the utterance ends. The controller filters duplicates and submits current revisions. Endpoint detection starts a new utterance identity without requiring another recording. Transcript output is separate from the eventual accepted movement decision.

**Verification:** the historical native WAV check produced 17 partials, one endpoint, expected phrases and no errors; it opened no microphone. Live speech accuracy/latency remain unmeasured. **Video:** 01:44-02:08.

## F04 - Natural-Language Movement Interpretation

**Reference/status:** Stage 1 F04; implemented with local SemIf/Qwen instead of the original provider assumption (D01/D13). **Design:** UC01/UC02; SD01/SD02. **Patterns:** Adapter, Command, MVC. **Type:** hybrid.

**Files, responsibilities and key methods:** VoiceCommandAgent.interpret() schedules the DecisionModel.decide() contract. OpenJevAdapter submits HTTP in `decide()` and validates the allowlisted output in `parseResponse()`. The SemIf bridge runs `DecisionEngine.decide()` with genuine Qwen logits. ControlManager.accept()/plan() creates commands; ActionExecutor.execute() applies them.

**Input -> collaboration -> result:** spoken text or typed `go up` reaches the same agent. Qwen chooses a bounded semantic action; Java stamps trusted identity, validates it and changes the game through commands. The view shows accepted type/directions or no-action/error feedback. The model never calls game methods. `--demo` explicitly selects a deterministic parser and is not the AI implementation.

**Verification:** recorded genuine-model and packaged CLI checks exercised the complete path. Finite samples do not guarantee every paraphrase. Current input errors stop controls; an unclear `NO_ACTION` leaves existing movement unchanged. **Video:** 03:44-04:30.

## F05 - Context-Aware Commands

**Reference/status:** Stage 1 F05; implemented and extended with repeat memory (D18). **Design:** UC03; SD03. **Patterns:** Adapter, Command. **Type:** hybrid.

**Files, responsibilities and key methods:** ControlContext stores held directions, prior accepted text and MovementIntent. VoiceCommandAgent.interpret() freezes one utterance's baseline; OpenJevAdapter.decide() sends it to Qwen. ControlManager.getContext()/accept()/plan() maintain/replay memory; ApplicationController.acceptanceMessage() explains missing history.

**Input -> collaboration -> result:** `other way` uses initially held directions; `keep going` preserves movement/timer. After timed movement expires, `again` asks Qwen for `REPEAT_LAST`; Java creates fresh commands from the stored direction/duration. With no history, repeat changes nothing and explains why. The model receives no map or goal.

**Verification:** frozen-context/history/no-history tests pass; eight recorded CPU repeat/history samples passed under a diagnostic deadline. Updated default-budget GPU verification is pending. **Video:** 04:35-05:37.

## F06 - Live Command Correction / Interruption

**Reference/status:** Stage 1 F06; implemented with D11/D12/D15/D16 refinements. **Design:** UC04; SD04. **Patterns:** Adapter, Command. **Type:** hybrid.

**Files, responsibilities and key methods:** ApplicationController.handleTranscript()/submitTextCommand() create newer versions. VoiceCommandAgent.interpret()/reset() bound work to two active calls and one latest pending input. ControlManager.beginInput()/accept()/plan() reject stale results and prepare replacements; ActionExecutor.cancelAll()/execute() replace active timed work.

**Input -> collaboration -> result:** `right - no, left` produces revised text. Qwen interprets the correction; only a matching version/epoch/control revision may act. Java cancels old timed work before applying replacement directions. A typed instruction supersedes remaining partials of an older spoken utterance. Cancellation cannot later release a newer command.

**Verification:** stale-result/replacement tests pass. The historical pre-repeat four-partial GPU burst returned LEFT after 990.7 ms within 2000 ms; this is not current repeat-schema timing evidence. The linked replacement example uses GUI typed text; detailed spoken-correction coverage remains unverified. **Video:** 04:35-05:37.

## F07 - Combined Directional Commands

**Reference/status:** Stage 1 F07; implemented. D13 describes axis readouts. **Design:** UC05; SD04. **Patterns:** Adapter, Command. **Type:** hybrid.

**Files, responsibilities and key methods:** DecisionEngine.decide() obtains model-selected semantic directions; OpenJevAdapter.decide() transmits the request and validates the returned choice. ControlManager.validate()/plan() enforce a compatible set. KeyDownCommand.execute() calls `GameModel.press()`; Player.refreshVelocity() normalizes speed.

**Input -> collaboration -> result:** `up and right` selects both axes. Java releases unwanted old inputs, presses the desired pair and displays diagonal movement/held directions. Opposing pairs are excluded from the provider vocabulary and rejected by Java validation.

**Verification:** compatibility/normalized-speed tests pass; recorded model samples covered all four perpendicular pairs. **Video:** 05:37-06:23.

## F08 - Magnitude / Timed Commands

**Reference/status:** Stage 1 F08; implemented with timed replay under D18. **Design:** UC05/UC03; SD04/SD03. **Patterns:** Adapter, Command. **Type:** hybrid.

**Files, responsibilities and key methods:** OpenJevAdapter.decide() limits semantic durations to 100/200/400 ms. ControlManager.validate()/plan()/advance() validate/create commands and change revision

at expiry. TimedPressCommand.execute()/advance()/cancel() own duration; ActionExecutor.advance() advances/removes timers.

**Input -> collaboration -> result:** `a little left` selects a 200 ms hold. The controller queries `ControlManager.secondsUntilExpiry()`/`ActionExecutor.secondsUntilExpiry()`, simulates each slice up to that boundary, then advances/releases the timer. This preserves full movement distance across frame sizes. Equivalent partials within that utterance cannot extend its timer. A fresh `again` creates a new timed command from remembered semantics. Timing uses the simulation loop, not independent delayed callbacks.

**Verification:** expiry, quantization, interruption and repeat tests pass. Packaged live CPU CLI replay used a 30000 ms diagnostic interpretation budget; the movement hold remained 200 ms of simulation time. **Video:** 06:23-07:41.

## F09 - Control-State and Conflict Management

**Reference/status:** Stage 1 F09; implemented with failure/replay refinements. **Design:** UC01-UC06; SD01/SD03/SD04/SD05. **Patterns:** Command. **Type:** deterministic.

**Files, responsibilities and key methods:** GameModel owns held inputs. ControlManager.beginInput()/accept()/validate()/reset() own version/epoch/revision and memory. ActionDecision.withRequest() receives local metadata from VoiceCommandAgent. ActionExecutor.cancelAll(), ReleaseAllCommand.execute() and ApplicationController.failCurrent() neutralize current failures.

**Input -> collaboration -> result:** each proposal is checked against current transcript/session/control revision before execution. Invalid current actions and current timeouts release controls; stale responses/failures cannot alter a newer session. Duplicate partials cannot rearm timers/repeats. Pause/fall/restart/current failure clear memory; ordinary stop, expiry and keep/no-action preserve it.

**Verification:** tests cover malformed/stale proposals, opposite directions, duplicate timers, expiry revision, reset/history and late failures. **Video:** 07:47-08:17.

## F10 - Fall Detection and Start Recovery

**Reference/status:** original F10 - Fall Detection and Checkpoint Recovery; modified and implemented. D17 removes checkpoint progression. **Design:** UC06; SD05. **Patterns:** State, Observer, Command. **Type:** deterministic.

**Files, responsibilities and key methods:** GameModel.evaluatePosition()/resetToStart() detect the unsafe footprint and restore `Level.start()`. FallingState.update() initiates reset; RespawningState.update() restores active play. ApplicationController.onGameEvent()/invalidateControls() invalidate AI work and controls; GameView.onGameEvent() presents lifecycle feedback.

**Input -> collaboration -> result:** human-directed movement leaves the corridor. FallingState and the fall event trigger epoch invalidation, interpretation/timer cancellation and release. Later lifecycle ticks restore start position and zero velocity, then ActiveState. If listening was requested, voice capture restarts; movement requires fresh input. This deliberately changes recovery from preserving checkpoint progress to losing all progress on a fall.

**Verification:** saved tests check repeated falls after level progress return to start and reject pre-fall responses. Specific real microphone restart/recovery coverage remains unverified. **Video:** 08:18-09:20.

# 3. Implemented design patterns

Maintained pattern explanations

The five patterns proposed in Stage 1 remain implemented: **MVC, Command, Adapter, Observer and State**. Each solves a concrete problem in the shared Java application. The change record explains implementation refinements; no pattern has been replaced and no repeated use is counted as an additional pattern.

## Current implementation UML

Diagram source: Implemented class diagram (embedded in the appendix).

The current class diagram is a UMLet-exported SVG. It shows the shared GUI/CLI architecture, control epoch/version/revision fields, `ActionExecutor.activeTimed`, `TimedPressCommand.remainingMs`, structured movement memory and the external SemIf boundary. Its historical "Stage 1 revised" title identifies the design lineage; the classes and key operations describe the current implementation. Concrete classes inherit interface operations, so inherited methods need not be repeated in each box.

The use-case diagram and sequences describe the implemented participants: SD01 voice, SD02 CLI, SD03 context/repeat, SD04 correction/timed actions, and SD05 start recovery. Original diagram sources/exports remain in Git history; they are not claimed to exist as current repository files.

**November 7 timing refinement:** SD04's final timer subsection still shows the earlier timer-before-simulation order. BF03 now queries the next expiry, simulates that bounded game slice, then advances/releases the timer. Re-export that subsection from UMLet before final submission. The class roles, timer ownership, other sequence flows and five patterns are unchanged; BF03 records the exact method changes.

## 1. MVC - shared behavior for two frontends

**Problem:** JavaFX and the terminal need different rendering/input mechanisms while exposing the same game and agent functionality. Duplicating movement, interpretation or recovery in the views would make behavior diverge.

| Participant and source | Role and actual methods |
|---|---|
| GameModel, Player, Level | Model: authoritative held inputs, player state and geometry. `update()`, `getSnapshot()`, `evaluatePosition()`, `move()`, `isSafe()`, `isGoal()`. |
| ApplicationController | Controller: serializes input, decision acceptance, simulation and cleanup. `submitTextCommand()`, `handleTranscript()`, `handleDecision()`, `advance()`, `onGameEvent()`. |
| GameView | Presentation contract: `render()`, `showTranscript()`, `showVoiceStatus()`, `showMessage()`. |
| JavaFxGameView, CliView | Views: draw snapshots and present feedback. GUI `submitTypedCommand()`/`requestStartVoice()` route intent to the controller; CLI `readCommand()` returns text for the launcher to submit. |
| Main | Composition: `compose()` builds the shared core for either frontend; `runCli()` handles terminal session commands. |

**How it follows the pattern:** a GUI text submission or CLI instruction reaches the same controller and agent. Accepted commands mutate `GameModel`; immutable snapshots reach the selected view. JavaFX scheduling and JLine terminal redraw remain presentation concerns. Neither renderer advances player physics or interprets direction language.

**Why useful / without it:** fixes to collision, correction or repeat behavior apply to both frontends. Without this separation, consistent GUI/CLI tests and alternate presentation would require changes to duplicated logic.

**Stage 1 comparison:** MVC is retained. D03 makes `GameView` inherit the observer contract, D04 adds GUI typed input, and D06/D19 refine rendering. These change presentation plumbing rather than replacing MVC. Features: F01-F04 and shared presentation of F05-F10.

## 2. Command - cancellable deterministic movement

**Problem:** a new correction, fall or timeout must interrupt old timed movement without a delayed release affecting a newer command. AI output must not directly mutate game state.

| Participant and source | Role and actual methods |
|---|---|
| GameCommand | Command interface: `execute(GameModel)` and `cancel(GameModel)`. |
| KeyDownCommand, KeyUpCommand | Concrete commands: press or release one direction through `execute()`; cancellation never reintroduces an already released input. |
| TimedPressCommand | Concrete timed command: `execute()`, `advance()`, `cancel()` own one compatible direction set and its remaining duration. |
| ReleaseAllCommand | Concrete neutralizing command: `execute()` clears held controls. |
| ControlManager | Client/planner: `accept()`, `validate()` and `plan()` translate a fresh semantic decision into command objects. |
| ActionExecutor | Invoker: `execute()`, `cancelAll()`, `advance()` own the active timed-command list and rollback partial failures. |
| GameModel | Receiver: `press()`, `release()`, `releaseAll()` update authoritative inputs. |

**How it follows the pattern:** a `SWITCH LEFT` proposal becomes release-right and press-left commands. The manager computes the desired set, cancels old timed work, then the executor applies the replacement on the application loop. Timer expiry is advanced on that same loop; there is no detached delayed-release callback. A `REPEAT_LAST` creates fresh commands from remembered semantics, never reuses an old command object or request identity.

**Why useful / without it:** cancellation, timed behavior and failure rollback are testable without an LLM. Without explicit command objects, these responsibilities would be spread across interpretation callbacks and game mutation code, making stale timer effects harder to prevent.

**Stage 1 comparison:** the planned pattern/classes remain. D10 specifies elapsed-time advancement; D11 centralizes invalidation; D18 adds fresh replay through the existing command mechanism. No general undo/history system is claimed. Features: F04-F10 where decisions, interruption or recovery require commands.

## 3. Adapter - isolate speech and decision providers

**Problem:** native sherpa-onnx APIs and SemIf's local HTTP/logits service have different input, lifetime and failure conventions. Those details must not spread into game/control logic.

| Participant and source | Role and actual methods |
|---|---|
| StreamingSpeechRecognizer | Target interface: `start(onPartial, onEndpoint, onError)`, `acceptAudio()`, `stop()`, `close()`. |
| SherpaOnnxAdapter | Adapter: creates the native online recognizer/stream, feeds audio, converts partials/endpoints and releases native resources. `start()`, `acceptAudio()`, `stop()`. |
| sherpa-onnx `OnlineRecognizer` / `OnlineStream` | External adaptees, used through the adapter's typed Java API. |
| DecisionModel | Target interface: asynchronous `decide(TranscriptUpdate, ControlContext)` returning `ActionDecision`. |
| OpenJevAdapter | Adapter: `decide()` builds a control-only request; `parseResponse()` checks backend/model and maps an allowed label to semantic output. |
| SemIf bridge and pinned Qwen3.5 4B | External adaptee boundary: `/decide` uses genuine SemIf logits scoring and returns a bounded choice. The bridge is supporting model-runtime code; the application remains Java. |

| Participant and source | Role and actual methods |
|---|---|
| VoiceInputService, VoiceCommandAgent | Clients depend on the target interfaces, allowing controlled substitutes in tests. |

**How it follows the pattern:** speech callbacks become provider-independent transcripts; HTTP/scoring output becomes a provider-independent semantic decision. Neither adapter calls `GameModel.press()` or performs route planning. The optional DemoDecisionModel implements the same target for explicitly labeled offline use; it is not the real LLM integration or an automatic fallback.

**Why useful / without it:** native deployment and provider protocol changes can be isolated and tested. Without adapters, the controller would need native recognition and HTTP/schema details, and testing failures would require live hardware/services.

**Stage 1 comparison:** the same two target interfaces/adapters remain. D01 corrects the provider identity to SemIf/Qwen, D13/D16 refine bounded readouts, and D14 specifies native deployment. These are adaptee/integration changes, not a new counted pattern. Features: F02-F08.

## 4. Observer - lifecycle cleanup and presentation

**Problem:** a fall or goal must notify both cleanup/control coordination and presentation without the game depending on a particular GUI or terminal class.

| Participant and source | Role and actual methods |
|---|---|
| GameModel | Subject: `addObserver()`, `removeObserver()`, `notifyObservers()` publish domain events synchronously. |
| GameObserver | Observer interface: `onGameEvent(GameEvent)`. |
| GameEvent, GameEventType | Event value/type: state change, fall, respawn and goal, with an immutable snapshot. |
| ApplicationController | Concrete observer: `start()` registers cleanup first; `onGameEvent()` invalidates controls/agent work and coordinates voice recovery. |
| GameView, JavaFxGameView, CliView | Presentation observers: inherited/default or overridden `onGameEvent()` renders snapshots and lifecycle messages; CLI throttles ordinary refresh while preserving lifecycle output. |

**How it follows the pattern:** `GameModel.evaluatePosition()` publishes `FALL_DETECTED`; the controller cancels old work, and the selected view presents the fall. Respawn/goal notifications follow the same contract. The game refers to `GameObserver`, not frontend implementations. Notification uses a copied subscriber list and isolates an observer's runtime exception so other subscribers still receive the event.

**Why useful / without it:** lifecycle cleanup and rendering can evolve independently. Without observer notification, the game would need direct controller/view calls or duplicated polling logic. Synchronous ordering here is deliberate, allowing cleanup before presentation observers.

**Stage 1 comparison:** the planned subject/observers remain. D03 centralizes the view contract, D08 adds removal/exception isolation, and D17 removes checkpoint events as part of the documented gameplay change. Features: F01/F10 and lifecycle control cleanup supporting F02/F09.

## 5. State - explicit player lifecycle

**Problem:** movement, recovery and goal completion need different behavior. Repeating lifecycle conditionals across game updates and frontends would make it easy to move during recovery or after finishing.

| Participant and source | Role and actual methods |
|---|---|
| Player | Context: stores `PlayerState`; `update()` delegates, `setState()` changes behavior and stops velocity when movement is disallowed. |
| PlayerState | State interface: `update(player, game, deltaTime)`, `allowsMovement()`, `status()`. |
| ActiveState | Concrete active state: subdivides movement and evaluates position in `update()`; permits movement. |

| Participant and source | Role and actual methods |
|---|---|
| FallingState | Concrete falling state: disallows movement; its next `update()` calls `GameModel.resetToStart()`. |
| RespawningState | Concrete recovery state: next `update()` restores `ActiveState` and publishes `RESPAWNED`. |
| FinishedState | Concrete completed state: disallows movement and keeps velocity zero in `update()`. |
| GameModel | Transition collaborator: `evaluatePosition()`, `resetToStart()`, `restart()` select lifecycle transitions. |

**How it follows the pattern:** the normal path is Active -> Falling -> Respawning -> Active, or Active -> Finished. These are different state objects with behavior, not only an enum with scattered branches. `PlayerStatus` is the immutable presentation value; it does not replace polymorphic state execution.

**Why useful / without it:** each lifecycle's update policy can be tested in isolation, and frontends receive the same status. Without State, movement and recovery rules would become repeated conditional checks and could disagree across paths.

**Stage 1 comparison:** all four planned states remain. D02 adds a typed status enum; D17 changes the recovery destination from a saved checkpoint to the level start. The State pattern still implements the same lifecycle separation. Features: F01/F10, with lifecycle gating used by F09.

## Verification boundary

The saved ordinary Java suite exercises game states, observer subscription, cancellation/rollback, freshness, both presentation contracts and provider boundaries. Recorded real-model and native-ASR checks are separate from controlled tests. The user has given general gameplay acceptance; specific microphone coverage and the updated default-budget GPU gate remain unverified. See validation for dates, commands and limitations, and the feature mapping for pending video references.

## Current UML and implementation contracts

The class model shows `ActionDecision`/`ControlContext` epoch and revision metadata, `ControlManager` freshness state, `ActionExecutor.activeTimed`, and the remaining duration owned by `TimedPressCommand`. These make concurrency and timer ownership structurally visible, as suggested by the Stage 1 feedback. F02/F03/F04 have separate acceptance criteria in sections 2 and 7.

| Diagram | Current UMLet SVG | PDF appendix |
|---|---|---|
| Class model | class-diagram.svg | A |
| Use cases UC01-UC06 | use-case-diagram.svg | B |
| SD01 - GUI voice command | sd01-gui-voice-command.svg | C |
| SD02 - CLI text command | sd02-cli-text-command.svg | D |
| SD03 - Contextual command and repeat | sd03-contextual-command.svg | E |
| SD04 - Correction, combined and timed actions | sd04-live-correction.svg | F |
| SD05 - Fall and start recovery | sd05-fall-recovery.svg | G |

# 4. Changes and rationale

The original Stage 1 snapshot is retained in Git history at `56da5fd95c43dde4760032e59a36787a2c74049c`. The revised design and current diagrams reflect the implemented local-model, repeat-memory, start-recovery, and terminal-display contracts. The following record explains changes against the earlier baseline rather than treating the revised design as proof that no changes occurred.

Maintained change record

This is the consolidated **Original design -> Implemented change -> Reason -> Impact** record required by the Course Project Stage 2 Instructions. F01-F10 remain traceable through the feature mapping. The same five patterns remain implemented; pattern roles and methods explain their structural realization.

## Baseline, current artifacts and evidence

The original Stage 1 design is preserved in Git history at commit `56da5fd95c43dde4760032e59a36787a2c74049c`. The revised Stage 1 report and current UMLet-exported SVGs describe the selected implementation. Earlier PlantUML sources, rendered previews and `docs/uml/stage1/` archives mentioned in development records are historical artifacts in Git history, not current-file promises. UMLet replaced the diagram authoring/export format; this did not change application behavior or add a design pattern.

Recorded verification progressed from 63 ordinary Java tests, to 73 after repeat support, to 76 after terminal redraw. Nine controlled bridge tests and 28 CPU semantic samples were recorded for repeat support. Earlier 20-case GPU and four-partial burst checks predate repeat. Historical diagram rendering/geometry/hash checks apply to the versions named in validation, not automatically to replacement SVGs.

The user reports having played the game and confirms that it works. The November 6 feature recording adds observed GUI microphone/transcript, text-control and CLI scenarios. Feature links and timestamps are supplied; YouTube returned playable status on November 7. Detailed device-failure coverage and the updated 2000 ms GPU burst check remain unverified. These limits do not imply that the corresponding code is missing.

## D01 - Correct the decision provider identity

**Original:** the initial report interpreted OpenJev as the unrelated hosted `openjev.sh` service with an `openjev` model alias.

**Implemented change:** retain Java `DecisionModel`/`OpenJevAdapter`, but connect to a project loopback HTTP bridge for genuine **SemIf (formerly OpenJev), using Qwen3.5 4B**. The upstream SemIf commit is pinned to `23cf1f39fc9534fe81437200959b6dfc7106e45a`. Python hosts the external inference dependency; the game, frontends and controls remain Java.

**Reason / original need:** correct the provider research error after the user identified the intended project and model. Natural-language movement interpretation remains the original user need; it now runs locally without an API key.

**Impact:** F04-F08; UC01-UC05; Adapter preserved. OpenJevAdapter.decide()/parseResponse(), DecisionModel.decide(), and DecisionEngine.decide()/load_scorer() implement the boundary. Genuine-model and packaged CLI checks are recorded separately from demo tests.

## D02 - Concrete immutable values and typed status

**Original:** value objects were conceptual and snapshot lifecycle status was string-valued.

**Implemented change:** use records for `Vector2`, `GameSnapshot`, `GameEvent` and agent/control values, plus `PlayerStatus`. Copy direction sets and level geometry into immutable collections.

**Reason / original need:** safely publish one model to both views and avoid spelling-dependent lifecycle handling. Movement and recovery behavior remain unchanged.

**Impact:** F01/F05/F09/F10; UC01-UC06; MVC, Observer and State preserved. GameSnapshot constructor, ControlContext constructor, Level constructor, Player.status(), PlayerStatus define the values. The enum is for presentation; `PlayerState` objects still execute lifecycle behavior.

### D03 - One presentation/observer contract

**Original:** both concrete views separately implemented `GameView` and `GameObserver`; transcript/status methods were frontend-specific.

**Implemented change:** `GameView` extends `GameObserver`, with default transcript/status hooks and lifecycle presentation. Concrete views provide rendering and messages; CLI overrides event handling for its output policy.

**Reason / original need:** give the controller one presentation contract and centralize common fall/respawn/goal messages without duplicating the simulation.

**Impact:** F01-F04/F10; UC01/UC02/UC06; MVC and Observer refined, not replaced. GameView.onGameEvent()/render()/showMessage(), CliView.onGameEvent(), ApplicationController.start() implement the change.

### D04 - Add GUI natural-language text input

**Original:** GUI instructions entered through speech; CLI instructions were typed.

**Implemented change:** add the GUI Text Instructions field and Send instruction action, routed to the same text-command controller method used by CLI.

**Reason / original need:** allow users to exercise/correct movement when audio setup is unavailable. This is an added entry route for F04, not direct keyboard movement or a substitute for F02/F03.

**Impact:** F04-F09; UC01/UC02/UC04; MVC preserved. JavaFxGameView.submitTypedCommand(), ApplicationController.submitTextCommand(), Main.runCli() share the pipeline. Historical GUI smoke testing exercised actual text-field movement with explicit demo mode.

### D05 - Add fresh-run and inspection controls

**Original:** session start/stop and checkpoint recovery were planned; a manual fresh-run control was not specified.

**Implemented change:** GUI Restart run and CLI `/restart`, `/state`, `/pause`, `/help`, `/quit` provide deterministic session operations. Restart invalidates pending work before resetting the game.

**Reason / original need:** let users restart the single level, inspect exact state and stop a session. These supporting operations are not counted toward the ten meaningful features.

**Impact:** F01/F02/F09/F10; UC01/UC02/UC06; MVC, Command, Observer and State preserved. ApplicationController.restart()/pauseControls(), GameModel.restart(), Main.runCli() and the GUI button handlers implement them. GUI and packaged CLI checks covered reset/release behavior.

### D06 - Implement responsive presentation scheduling

**Original:** snapshots and throttled CLI display were described without concrete scheduling utilities.

**Implemented change:** JavaFX coalesces snapshots/transcript/messages on its application thread. The initial CLI used a 61 x 27 ASCII overview and approximately one-second refresh; explicit quiet mode suppressed periodic scripted output. D19 later replaces the scrolling interactive presentation. The current GUI also highlights the neutral player at the start; this is presentation of existing state.

**Reason / original need:** prevent repaint accumulation and make the player/state readable without adding game mechanics. At the initial stage, Gradle-forwarded streams made `System.console()` alone insufficient to identify a usable terminal.

**Impact:** F01/F03/F04/F10; UC01/UC02/UC06; MVC and Observer preserved. JavaFxGameView.render()/scheduleUpdate()/drainUpdates(), CliView.formatSnapshot()/onGameEvent() are relevant. D19 supersedes the original append-only CLI refresh policy, not the shared model.

### D07 - Add an explicitly selected offline demo model

**Original:** semantic interpretation used the configured real `DecisionModel`.

**Implemented change:** add `DemoDecisionModel`, a limited deterministic parser selected only by explicit `--demo`. Real SemIf remains the default; there is no automatic fallback to demo when inference fails.

**Reason / original need:** permit application development/demonstration without loading an AI service. This does not replace the required LLM integration or establish AI accuracy.

**Impact:** F04-F09 development/demo path; UC01-UC05; Adapter target contract preserved. DemoDecisionModel.decide()/parse(), Main.compose(), SaySwearApplication.start() select/label the mode. Demo results are identified separately in validation.

## D08 - Define observer removal and failure isolation

**Original:** subject notification was planned without a subscriber-removal or exception policy.

**Implemented change:** notify a copied subscriber list, allow removal, and catch/log individual subscriber failures while continuing notification. Register controller cleanup before the presentation observer.

**Reason / original need:** a presentation failure must not prevent lifecycle cleanup; close/unsubscribe must be safe during notification.

**Impact:** F01/F09/F10; UC01/UC02/UC06; Observer refined. GameModel.removeObserver()/notifyObservers(), ApplicationController.start()/close() implement the policy. Observer removal and fall cleanup have deterministic tests.

## D09 - Choose a reproducible Java build

**Original:** Java/JavaFX were required without concrete build and dependency versions.

**Implemented change:** use Java 21, JavaFX 21, Gradle wrapper 8.14.2, pinned sherpa Java/native dependencies and JUnit. Platform classifiers select native dependencies for the host.

**Reason / original need:** provide reproducible compilation, ordinary tests and packaged GUI/CLI startup rather than requiring manually assembled classpaths.

**Impact:** all features and UC01-UC06; all five patterns retained, no added pattern. build.gradle defines `test`, `installDist`, `guiSmokeTest`, `asrSmokeTest` and `liveSemIfTest`; wrapper configuration pins Gradle. Historical suites increased 63 -> 73 -> 76 ordinary tests as functionality evolved; special tasks remain opt-in. D20 records Linux setup entry points.

## D10 - Specify elapsed-time simulation and timer ownership

**Original:** a fixed 60 Hz timestep and roughly one-tick timer quantization were proposed.

**Implemented change:** schedule updates every 16 ms, measure elapsed time and cap one update at 50 ms. Advance timers before game movement; subdivide movement for corridor collision checks. `ActionExecutor` owns active timed commands and `TimedPressCommand` owns remaining duration.

**Reason / original need:** bound motion after a delayed loop turn and prevent a large update from skipping across void. Timer tolerance follows the actual capped update; prolonged stalls can make simulation lag wall time.

**Impact:** F01/F08/F09/F10; UC01/UC02/UC05/UC06; Command and State preserved. ApplicationController.tickFromClock()/advance(), ActionExecutor.advance(), TimedPressCommand.advance(), ActiveState.update() implement this. Movement, quantization and collision tests cover deterministic behavior.

## D11 - Centralize request invalidation and cleanup

**Original:** sequences used explanatory operations `handleInputFailure(message, epoch, version)` and `invalidateControls(reason)`.

**Implemented change:** completion callbacks check captured identity and use private `failCurrent(message)`/`invalidateControls()` helpers. Explicit `close()` lifecycles release resources. Control reset advances the epoch before cancellation callbacks run; decisions carry local version/epoch/revision metadata.

**Reason / original need:** use one neutral cleanup path and prevent old successes/failures from changing a newer input session. Public implementation operations differ from the conceptual signatures without changing their responsibility.

**Impact:** F02/F04/F06/F09/F10; UC01/UC02/UC04/UC06; MVC, Command, Adapter and Observer preserved. ApplicationController.isCurrent()/failCurrent()/invalidateControls()/close(), ControlManager.reset()/accept(), VoiceCommandAgent.reset()/close(), VoiceInputService.stop()/close() implement cleanup. Tests cover stale

callbacks and bounded/reset behavior.

## D12 - Define precedence between voice and typed input

**Original:** the selected frontend supplied voice or typed text, so simultaneous input-source precedence was unspecified.

**Implemented change:** a typed command supersedes remaining partials of the current spoken utterance. Fresh voice becomes eligible after a new endpoint. Both sources share one version counter and control pipeline.

**Reason / original need:** prevent an older evolving transcript from undoing a newer typed correction after D04 added GUI text entry.

**Impact:** F02/F03/F06/F09; UC01/UC02/UC04; MVC and Adapter preserved. ApplicationController.submitTextCommand()/handleTranscript() and VoiceInputService.endpoint() implement source/utterance identity. Specific mixed microphone/text coverage remains unverified.

## D13 - Factor bounded semantic selection into model readouts

**Original:** the adapter conceptually obtained one bounded action from the provider.

**Implemented change:** genuine upstream full-prompt SemIf scoring selects action type, vertical `UP/DOWN/NONE`, horizontal `LEFT/RIGHT/NONE`, and an optional duration. One to four readouts compose one allowed action. The original seven-type/51-label version became eight types/52 labels with D18.

**Reason / original need:** respect the provider's per-readout option limit and resolve observed direction-set errors while supporting every diagonal. Each axis receives the transcript and its initial held direction; no map, position or destination is included.

**Impact:** F04-F08; UC01-UC05; Adapter preserved, deterministic Command boundary unchanged. DecisionEngine.decide()/validate(), OpenJevAdapter.createChoices()/parseResponse() enforce the vocabulary. Historical model failures, eventual sampled successes and controlled bridge tests remain recorded in validation; this is not a universal accuracy claim.

## D14 - Specify compatible native model deployment

**Original:** native model deployment and quantization were unspecified.

**Implemented change:** use pinned Q4_K_M GGUF weights/reference tokenizer. The Windows deployment uses a project-local overlay pairing the same-version official portable AVX2 CPU DLL with unchanged CUDA libraries; the installed dependency is preserved. Explicit GPU allocation extends the upstream CPU allocation setting while retaining its scoring.

**Reason / original need:** the initial CUDA wheel's CPU library required AVX512 unavailable on the verification CPU. GPU offload still needs compatible CPU instructions. Quantization makes local inference practical within the available memory.

**Impact:** F04-F08 runtime; UC01-UC05; Adapter boundary preserved. configure-semif-native.py `configure()`, server `load_scorer()`, SemIf setup document provenance and startup. Historical readiness/model checks passed with 34/34 layers offloaded; this is platform-specific evidence.

## D15 - Adjust the decision deadline

**Original:** the initial implementation used a 1000 ms deadline including queue time.

**Implemented change:** default to 2000 ms while retaining finite provider timeouts, two active calls, one replaceable latest pending input and fail-closed current failures. Explicit launch options override environment settings.

**Reason / original need:** the initial four-partial burst expired at 1000 ms. A later pre-repeat burst returned the latest correction after 990.7 ms with the 2000 ms budget; the larger allowance accommodates queue/runtime variation without adding artificial delay.

**Impact:** F04/F05/F06/F08/F09; UC01-UC05; Adapter/Command unchanged. VoiceCommandAgent.interpret()/expire(), OpenJevAdapter.decide(), AppConfig.parse(), run.ps1 implement deadlines/configuration. Pre-repeat real CLI/burst checks passed; current repeat-schema GPU budget

verification remains outstanding.

## D16 - Normalize correction word boundaries for axis readouts

**Original:** the same raw transcript representation reached every model readout.

**Implemented change:** preserve raw text in Java, the HTTP request and action-type scoring. Only after Qwen selects `SWITCH`, split commas/whitespace and join words with `,` for axis scoring.

**Reason / original need:** the unchanged ASR-like correction burst exposed sensitivity to unpunctuated text. Generic, idempotent formatting supplies word boundaries; it does not identify direction words or parse correction intent. Qwen still selects each axis.

**Impact:** F06, with F05/F07 regression coverage; UC03/UC04/UC05; Adapter preserved. DecisionEngine.decide(), bridge tests, LiveSemIfTest record the change. Twenty isolated cases and the original four-partial burst passed together before repeat support; wider language robustness remains unestablished.

## D17 - Replace checkpoint recovery with start recovery

**Original:** F10/UC06/SD05 included checkpoint activation and respawn at the latest checkpoint. `Level`/`GameSnapshot` exposed progress, `resetToCheckpoint()` restored it, and views showed checkpoints.

**Implemented change:** remove checkpoint geometry, saved positions, activation events and progress displays. F10 becomes **Fall Detection and Start Recovery**, retaining its ID. Every fall restores `Level.start()` with zero velocity and no held inputs, while preserving cancellation/freshness and the fall/respawn lifecycle.

**Reason / original need and scope:** the user requested that every fall lose all progress, matching the intended rage-game experience. Recovery still allows another attempt; preserving intermediate progress is deliberately removed, not claimed as implemented.

**Impact:** F10 and F01/F09 feedback/cleanup; UC06; State, Observer, Command and MVC preserved. GameModel.evaluatePosition()/resetToStart(), Level.start(), FallingState.update(), RespawningState.update(), ApplicationController.onGameEvent(), GameView.onGameEvent() implement recovery. `GameSnapshot`/`GameEventType` no longer contain checkpoint data/events. Current UC06/SD05 show the replacement; 63-test, GUI and CLI regressions passed at this change.

## D18 - Add structured repeat memory and refresh UML

**Original:** context contained held directions and prior accepted text, but no structured movement history. Seven action types/51 labels could not explicitly request replay after a timed movement expired.

**Implemented change:** add `MovementIntent(type, directions, durationMs)` for accepted PRESS/SWITCH/TIMED_PRESS and expose nullable `ControlContext.lastMovement`. Add empty-direction, zero-duration `REPEAT_LAST` as the eighth type/52nd label. Java creates new commands from remembered semantics; absent history produces neutral feedback. Permit one repeat per utterance, including across intervening partial interpretations.

**Reason / original need:** the user requested `again`. Resolved memory preserves exact duration/direction after expiry without asking the model to reconstruct an old action. Ordinary stop/expiry/keep/no-action preserve memory; pause/fall/restart/current failure clear it. This extends F05/F08/F09 rather than inventing another counted feature.

**Impact:** F05/F08/F09; UC03/UC05, with reset behavior in UC06; Adapter and Command preserved. MovementIntent, ControlContext, ControlManager.accept()/plan()/reset(), OpenJevAdapter.decide(), DecisionEngine.decide()/validate(), ApplicationController.acceptanceMessage() implement repeat. Restart an older running bridge to load the updated schema.

**Historical evidence/artifacts:** 73 Java tests, nine bridge tests, 20 unchanged baseline plus eight repeat/history CPU samples and packaged demo/live CLI checks passed. CPU samples used 60000 ms and live CPU CLI 30000 ms diagnostic caps, not the 2000 ms gameplay gate. At this stage original diagrams were archived under `docs/uml/stage1/` and 14 archived files were hash-checked while current diagrams were regenerated. Those archives/previews are now historical; current diagrams are UMLet SVGs.

## D19 - Keep the CLI display in place while editing input

**Original:** full ASCII frames printed about once per second, scrolling the terminal while the user typed; Windows launched CLI through Gradle-forwarded streams.

**Implemented change:** use JLine 3.30.16 `Status` for a fixed bottom panel and `LineReader` for editable input. Resize the map, include the JNI terminal provider and launch the installed distribution directly after `installDist`. Quiet, redirected and unsupported-terminal sessions use plain nonperiodic state/event output.

**Reason / original need:** the user requested a stable game display that preserves partial commands. Direct startup exposes actual terminal capabilities. Renderers continue consuming shared snapshots; no second simulation or interpreter is added. This supersedes D06's scrolling interactive output.

**Impact:** F01/F04 presentation and F09/F10 feedback; UC02, with UC06 recovery display; MVC and Observer preserved. CliView.open()/readCommand()/refreshDashboard()/updateDashboard()/close(), Main.runCli(), build.gradle, run.ps1 implement the change. Seven CLI tests, full 76-test suite, packaged demo, automatic piped fallback, actual PowerShell quiet launch and native Windows ConPTY checks passed. Specific native Linux terminal interaction remains unverified.

**UML impact:** class/SD02 show terminal ownership, redraw, editable input, fallback and cleanup; SD05 distinguishes ordinary throttled refresh from lifecycle events. Earlier regeneration/checks are historical; current corresponding SVGs retain those responsibilities.

## D20 - Add Linux launch and model-setup entry points

**Original:** recorded setup/launch automation was primarily PowerShell; portable Java source alone did not provide equivalent Linux bootstrap commands.

**Implemented change:** add run.sh, setup-semif.sh, run-semif.sh and setup-asr.sh. These locate the repository, honor documented options/environment and provide Java launch, explicit CPU/GPU model setup, and pinned ASR download/extraction. The ASR script uses `verify_archive()` for size/hash checks and an extraction-completion marker so failed extraction is repaired on rerun.

**Reason / original need:** support users building/running on Linux without translating Windows-specific paths, virtual-environment layout and native DLL handling. This is a supporting deployment change, not a new game feature or a rewrite of the Java application.

**Impact:** setup for F01-F10; UC01/UC02 deployment prerequisite; no pattern added or replaced. ASR setup, SemIf setup, AppConfig.parse() document configuration. Recorded Git Bash/fixture checks verify selected script control flow, including interrupted extraction; they do not establish native Linux inference, audio or terminal behavior. Consult validation for exact executed scope.

## Preserved responsibilities and remaining completion work

The human chooses every direction and turn. AI receives control context, never level geometry or a destination, and returns only a bounded semantic proposal. Java owns schema/freshness validation, memory replay, cancellation, timing, simulation, fall detection and recovery. Both frontends share that core.

No design pattern has been replaced: MVC, Command, Adapter, Observer and State retain their roles with the refinements above. No checkpoint functionality is silently claimed after its removal. All features have timestamped video links; playable metadata and matching duration were checked November 7. The recording shows GUI microphone behavior; detailed device-failure coverage and current default-budget GPU verification remain unverified. Full Stage 3 testing/KUMA evaluation is a separate deliverable.

## November 7 defect refinements

The BF01-BF05 record provides Before -> Change -> Verification -> Review for five separate fixes. BF01/BF02 repair launcher status/help. BF03 corrects F01/F08/F09 timed execution through existing controller/Command participants. BF04 repairs F02 capture cancellation within the existing Adapter boundary. BF05 improves F01 CLI presentation through the existing MVC/Observer participants. No feature or pattern is added or removed. SD04's final tick subsection needs the BF03 ordering refinement; the UMLet SVG is retained for that manual update.

# 5. AI tools and collaboration log

The tool/model/task/accounting table and current human contribution below are part of the maintained collaboration log. Codex usage and subscription cost are student estimates, not measured platform exports. Local token and compute totals are identified as unavailable rather than assumed zero.

This is a representative Stage 2 record, updated for `Project-Stage2-Instruction.pdf` and the baseline project instructions. It records actual instructions, generated work, human decisions, and corrections. General gameplay acceptance and detailed source/test review are recorded separately.

**Current status - November 7, 2026 (Toronto):** the user reports having played the game and confirms that it works properly. The original statement supports general gameplay acceptance. The November 6 recording index now identifies observed GUI microphone/text and CLI scenarios, with YouTube playable metadata and matching duration confirmed November 7. Detailed source/test review remains unreported. The November 7 ordinary suite has **92 passing tests**; `build installDist` passed after the five defect fixes linked below. The pre-repeat 990.7 ms GPU burst remains historical evidence; the updated 2000 ms repeat-enabled burst has not been verified. The instructor accepted existing early commit history for weekly-progress evidence. Current report preparation and remaining evidence are tracked in validation. Earlier entries retain the evidence and pending status known when each interaction occurred.

## November 7, 2026 - Reproduced defects and separate fixes

- **Task/Input:** correct five verified Stage 2 defects and verify their behavior with regression tests.
- **AI Contribution:** reproduced and corrected shell exit/help defects, frame-dependent timed distance, a microphone startup/stop race, and slow CLI refresh. Used focused investigations and regression tests, with one code/test commit per fix. Updated the fix record, mappings and verification evidence. Temporary diagnostics remain local.
- **Human Contribution:** established the gameplay/control requirements and supplied the gameplay recording that identified slow CLI feedback. Earlier gameplay acceptance remains recorded. Detailed source review has not been claimed complete.
- **Outcome:** 92 ordinary Java tests pass without failures/errors/skips; build and installed distribution pass. Packaged before/after probes confirm corrected exit status, help and timed distance. No real-model or microphone hardware tests were rerun for this audit. SD04 needs a UMLet timing-order refinement. The five code/test fixes were pushed to public main on November 7; the weekly summary links each commit.

## Tools and models

| Tool/platform | LLM/model | Features, classes, or development tasks | Total token cost/usage | Total money cost |
|---|---|---|---|---|
| OpenAI Codex, including delegated coding/review agents | GPT-6 model family, as identified by the session configuration; precise served snapshot unavailable | F01-F10; game/control core, JavaFX/CLI, controller/agent, adapters, concurrency review, tests, setup, diagrams, and implementation records | **Approximately 6 billion tokens** - student's estimate for project work, not a platform usage export | **Approximately CAD $282.50** - student-reported CAD $250 subscription charge multiplied by 1.13 for tax; project-specific allocation unmeasured |
| Local SemIf decision runtime | Qwen3.5 4B, Q4_K_M | Application interpretation for F04-F08, genuine model/agent checks, and packaged live CLI validation; `OpenJevAdapter` and `tools/semif/server.py` | **Unavailable** - cumulative local inference tokens were not recorded | No separately billed model API charge; local electricity and hardware costs were not measured |
| sherpa-onnx streaming recognition runtime | English streaming Zipformer; sherpa-onnx 1.13.5 | F03; `SherpaOnnxAdapter`, streaming audio integration, and native sample-audio validation | **N/A for LLM tokens** - audio recognition, with no LLM token billing counter | No separately billed ASR API charge; local electricity and hardware costs were not measured |

The October 27 Codex figures are student-supplied approximations. The 1.13 multiplier is treated as 13% tax, and CAD is assumed from the student's Toronto context; the billing period and whether the subscription also covered other work have not been specified. The CAD $282.50 figure is reported subscription spending, not a measured cost allocated solely to Say Swear. Confirm the currency, period, and estimate against available account records before submission. No private billing documents or account identifiers are included here. Local model token totals and compute costs cannot be inferred from test counts.

The AI models running **inside Say Swear** are distinct from the development assistant. The user selected **Qwen3.5 4B** and identified the intended backend as **SemIf**. The assistant then researched openjev.com and TheoLeeCJ/SemIf-OpenJev. Recorded checks cover native sample-audio ASR, genuine model cases, an earlier four-partial agent burst, and packaged live CLI interaction. Their dates, diagnostic budgets, historical failures, and remaining evidence limits appear in validation.

## October 20, 2026 - Authorized report update and gameplay acceptance

- **Task:** Bring Stage 2 documentation and the implementation report into line with the new Stage 2 instructions.
- **Input/Instruction:** The user authorized report/documentation work and stated, "I also played the game already so I know it works properly."
- **AI Contribution:** Audited the new requirements; separated existing implementation evidence from missing report, video, usage/cost, weekly-activity, and review records; prepared updated documentation and identified the remaining review items.
- **Human Contribution:** Reported personal gameplay experience and general acceptance that the game works properly. The user did not identify a particular frontend, microphone session, scenario coverage, or completed source-code review in this statement.
- **Outcome:** General gameplay acceptance is recorded. Existing test evidence remains unchanged. At the time, detailed review items, feature-video links, usage/cost totals, and weekly-progress evidence remained pending. The instructor's October 27 response about the early commit history is recorded in validation. This documentation update did not claim new application or model test results.

## Representative implementation interactions

| Interaction | Human contribution | AI contribution and actual result |
|---|---|---|
| C01 - Shared Java design | Specified small scope, both frontends, and human navigation; authorized implementation. | Implemented game/control, adapters, frontends, and controller coordination; recorded build and integration checks with their actual scope. |
| C02/C08-C11 - Correct and validate the backend | Selected Qwen3.5 4B and clarified that the intended backend was SemIf. | Corrected the earlier provider assumption, isolated native-runtime failures, factored bounded model choices, and recorded real inference/CLI cases separately from deterministic checks. |
| C04/C09 - Freshness and deadlines | Required live correction and interruption without stale actions. | Added epoch/version/revision validation, cancellation, bounded requests, and queue-inclusive expiry. Historical burst timing is preserved without claiming current repeat-enabled performance. |
| C12 - Start recovery | Rejected checkpoints because they conflicted with the rage-game concept. | Removed checkpoint progression and changed F10, state transitions, tests, views, and traceability to neutral start recovery. |
| C13 - Start marker | Requested a red arrow with white YOU ARE HERE text. | Added the presentation cue at the neutral starting position; no additional gameplay feature was counted. |
| C14 - Repeat after timer expiry | Identified that replaying the latest release action could make `again` ineffective; requested repeat support and diagram readability fixes. | Introduced `MovementIntent` and `REPEAT_LAST`; retained movement history after expiry, prevented duplicate same-utterance replay, and recorded repeat checks and remaining timing limits. |
| C15-C16 - Terminal display and diagrams | Reported scrolling CLI output and requested diagram consistency. | Added JLine in-place display and editable input, direct launch, plain fallback, terminal tests, and corresponding diagram changes. |
| C17 - Linux ASR setup | Identified the missing Linux setup script and requested it. | Added pinned/resumable/verified setup with fixture checks; recorded that native Linux recognition was not exercised. |

These representative interactions document actual scope decisions, corrections, generated work, and checks. The full log retains the original Task, Input/Instruction, AI Contribution, Human Contribution, and Outcome fields for C01-C17. The user's current gameplay acceptance is recorded above; it does not imply that every generated source file or test has received detailed human review.

# 6. Development and verification summary

## Current acceptance and weekly activity

This record separates executed checks from outstanding evidence, as of **November 7, 2026 (Toronto)**. Earlier development checks were recorded on September 30; later verification and setup work are dated below. On **November 7, 2026**, `build installDist` passed with **92 tests, zero failures, errors or skips** after five reproduced defect fixes. Earlier October 19 results (76 tests) remain historical evidence. The feature mapping identifies the corresponding implementation files.

## Current acceptance and verification status

- **General gameplay accepted:** on October 20, the user stated, "I also played the game already so I know it works properly." This is user-reported gameplay acceptance. The frontend, input device, scenarios exercised, and detailed source/test review were not specified.
- **Ordinary Java verification:** November 7 `build installDist` passes with 92 tests. The defect record explains five independently committed fixes and their regressions. Packaged probes also confirm failure exit 1, usable help with an invalid timeout, and exact 0.70/1.40-unit distances for 100/200 ms holds across 10/50 ms frames.
- **Real-model timing:** the **990.7 ms** GPU correction burst below predates repeat support. The updated repeat-enabled **2000 ms** GPU burst gate remains unverified. Repeat CPU cases used longer diagnostic deadlines, as recorded below.
- **Recorded feature demonstration:** the November 6 recording covers F01-F10 through GUI microphone/text and CLI gameplay. On November 7, YouTube returned player status OK without account credentials and duration 691 seconds, matching the reviewed source timeline. This is finite functional evidence; it does not establish measured accuracy or latency.
- **Evidence still needed:** detailed student source review, confirmation of the AI accounting period/currency/estimate, and the SD04 UMLet timing-order refinement under BF03. The instructor accepted early commit history as recorded below. Detailed device-failure, native Linux and current GPU performance coverage remain limited.

## Weekly code-activity summary

The written Stage 2 instructions set a baseline of **five meaningful implementation/test commits pushed during each Toronto-dated reporting period**. On October 27, the student supplied the instructor's reply to a direct question about substantial implementation completed before October 20 and the weekly count: the existing history works. The screenshot remains outside the repository because it contains names and account details. This report presents the actual history as the accepted evidence; it does not assign early commits to later weeks or claim that each week's numerical baseline was met.

| Period | Dates, inclusive (2026, Toronto) | Written baseline | Actual code/test pushes as of November 7 | Progress evidence |
|---|---|---|---|---|
| 1 | October 20-26 | 5 | 0 | Early history accepted; links below. |
| 2 | October 27-November 2 | 5 | 0 | Early history accepted; links below. |
| 3 | November 3-9 | 5 | 5 | Five code/test fixes pushed November 7; links below. |

The early Stage 2 evidence includes the shared game, agent, GUI/CLI implementation and tests (September 30), Linux application and SemIf launchers (October 6), and the integration into public main with Linux ASR

setup (October 19). These are cited at their real dates. Report-only edits and artificial subdivisions are not represented as code progress.

Five meaningful code/test fixes were committed and pushed to public main on November 7:

| Commit | Development progress |
|---|---|
| d825af6 | Startup errors return failure status; subprocess exit regressions. |
| 6f270dc | Help remains available with invalid configuration; subprocess regressions. |
| 1d4a443 | Timed movement preserves full duration across frame sizes; distance, repeat and fall regressions. |
| 4cc51d8 | Cancelled native microphone startup cannot leak capture; controlled lifecycle regression. |
| 539368a | Interactive CLI refreshes during precision movement; terminal screen/input regression. |

Documentation-only commits are excluded from this code/test count. The first two reporting periods remain described by the instructor's acceptance of early history; November 7 publication does not change those earlier dates.

## Executed verification and limits

| Check | Recorded outcome | What it establishes / limit |
|---|---|---|
| Ordinary Java suite and distribution | November 7: 92 passing tests, zero failures/errors/skips; build/installDist passed after BF01-BF05. | Domain/control, freshness/cancellation, agent scheduling, mock HTTP, capture lifecycle, and CLI behavior. Does not run a real LLM or microphone. |
| SemIf bridge contracts | Nine controlled-score tests passed after repeat support. | Request/schema/action composition and structured memory. Does not measure real-model language accuracy. |
| Native ASR sample | Real 6.63-second WAV yielded 17 changing partials and one endpoint, expected phrases, zero errors. | Actual sherpa-onnx native/model streaming recognition; no microphone was opened. |
| Earlier genuine-model checks | Twenty baseline cases passed; isolated requests 242.2-503.1 ms with a 15-second diagnostic deadline; four-partial burst returned latest LEFT correction after 990.7 ms within 2000 ms. | Historical pre-repeat semantic/agent/CLI evidence. Not a measurement of current repeat-enabled or microphone latency. |
| Repeat-enabled CPU checks | Twenty baseline plus eight repeat/history cases passed with a 60000 ms diagnostic cap; packaged real CLI used 30000 ms. | Genuine Java/SemIf/Qwen repeat semantics and execution. Does not establish the default 2000 ms performance gate. |
| GUI and packaged CLI | Recorded JavaFX text-entry, movement, Stop & release, restart, packaged real/demo CLI, native Windows terminal, and piped fallback checks passed. | Actual application wiring and terminal behavior; demo and genuine-model results remain distinguished. |
| Linux ASR setup | Bash/GNU fixture checks passed on Windows: fresh/cached, resume, invalid archive, unsafe path, and interrupted extraction cases. | Downloader/control flow. Native Linux recognition, GUI, and interactive terminal behavior remain unverified. |
| User gameplay and feature recording | November 6 local recording reviewed; YouTube link and feature timestamps supplied. | Observed GUI microphone/transcript, text interpretation, contextual replacement, combined/timed/repeat actions, cleanup/fall and CLI play. No measured accuracy/latency guarantee or completed source review is inferred. |

Full commands, earlier failures, correction history, counts, diagnostic settings, and limits remain in the validation record. The November 7 ordinary suite was rerun for the defect fixes; live-model and microphone-hardware checks were not repeated. Report generation checks document structure, links,

diagram embedding, and rendered page layout separately from application verification.

## Important defects found and corrected

- **Incorrect provider identity:** replaced the unrelated hosted-service assumption with the user-selected local SemIf/Qwen runtime behind the same Java interface.
- **Native inference startup:** isolated an incompatible native instruction requirement and documented pinned compatible runtime setup without changing the application core.
- **Model direction/correction failures:** factored action/axes/duration readouts and used generic word-boundary formatting only for model-selected SWITCH axes; Qwen still chooses intent and directions.
- **Burst deadline failure:** increased the original queue-inclusive 1000 ms budget to 2000 ms after the unchanged correction burst expired; the historical pass and outstanding current repeat-enabled check remain distinct.
- **Repeat after automatic release:** introduced structured movement memory so expiry does not replace remembered movement with release, and same-utterance partials cannot rearm it repeatedly.
- **Scrolling CLI input:** replaced repeated full-screen printing with a fixed game panel and separate editable prompt, while retaining plain output for redirected/unsupported terminals.
- **Interrupted ASR extraction:** added a verified completion marker so a failed extraction is repaired rather than treating truncated files as ready.

## November 7 reproduced defect fixes

Detailed fix and regression record. Each row is one separate code/test commit pushed November 7; publication links appear in the weekly summary above.

| Fix | Corrected behavior | Regression evidence |
|---|---|---|
| BF01 | Failed startup returns a nonzero shell status. | Invalid subprocess invocations fail; help and CLI quit succeed. |
| BF02 | Help works with broken runtime settings. | Both help aliases and invalid timeout/environment combinations. |
| BF03 | Timed input moves for its full duration. | Nine duration/frame combinations, exact expiry snapshot, full repeat distance and fall cleanup. |
| BF04 | Cancelled native microphone startup cannot leak capture. | Delayed-open/stop race leaves the fake native line closed and not recording. |
| BF05 | Interactive CLI feedback refreshes at up to 10 Hz. | Independent terminal emulator sees motion at 250 ms while preserving partial input. |

SD04's final tick subsection needs a UMLet update: simulate a game slice bounded by the next timer expiry, then advance/release that timer. Existing class/pattern ownership remains accurate. The pre-fix feature recording still demonstrates the same command semantics.

## Focused human review and remaining submission work

The completed gameplay session is already accepted. Detailed source/test review has not yet been reported complete. Record its actual scope, corrections, and outcome in the collaboration log, including understanding of the shared core, the AI/execution boundary, freshness/cancellation, timer/repeat semantics, lifecycle recovery, and five pattern roles.

Video links and matching YouTube duration are confirmed. Remaining submission items are detailed source/test review, confirmation of the reported AI accounting, and the SD04 UMLet timing-order update. The instructor accepted early history; five November 7 fixes meet the period 3 code/test push count. Native Linux runtime, detailed device-failure coverage and current repeat-enabled GPU timing remain unverified.

# 7. Demonstration index

**Recording supplied November 6, 2026.** Say Swear feature demonstration covers the ten features and both interfaces. The local 11:30 recording was reviewed through sampled frames, closer inspection of key action transitions, and a local speech-recognition transcript. The rows describe the observed demonstration, rather than an unexecuted recording plan.

**Playback metadata checked November 7:** YouTube returned player status OK without account credentials and a duration of 691 seconds, matching the local recording reviewed for these timecodes. The November 6 processing condition has cleared. The source demonstration predates the November 7 fixes to precise timed distance and CLI refresh; those changes preserve the demonstrated commands and feature scope.

| Feature | Demonstration shown | Visible result and scope | Video URL | Time (start-end) |
|---|---|---|---|---|
| F01 - Real-Time Precision Movement | GUI movement during spoken instructions; CLI movement at 10:18-11:30. | Player position changes in the displayed level and CLI map/state. | Watch | 01:05-03:25 |
| F02 - Continuous Voice-Control Session | GUI Start listening, successive spoken instructions, and Stop & release. | One listening session accepts multiple utterances; listening and control status are visible. | Watch | 01:23-03:25 |
| F03 - Streaming Speech Transcription | GUI microphone input with the transcript visible during speech. | Partial text changes while the speaker continues; transcription is distinct from action selection. | Watch | 01:44-02:08 |
| F04 - Natural-Language Movement Interpretation | GUI text instructions: right, stop, and hello, using the displayed local SemIf/Qwen backend. | Directional movement, release, and unsupported-input no-action feedback. CLI use is shown separately at 10:18. | Watch | 03:44-04:30 |
| F05 - Context-Aware Commands | GUI right followed by other way; remembered timed repeats also appear in F08. | RIGHT is replaced by LEFT using current context; later repeats use stored movement. | Watch | 04:35-05:37 |
| F06 - Live Command Correction / Interruption | Shared F05/F06 segment: a newer other way instruction replaces active right movement. | The held direction and velocity reverse. This shows active replacement, not an artificial out-of-order response test. | Watch | 04:35-05:37 |
| F07 - Combined Directional Commands | GUI text: a little up and right. | The accepted timed action contains UP and RIGHT; both position coordinates change. | Watch | 05:37-06:23 |
| F08 - Magnitude / Timed Commands | GUI timed movement followed by fresh again instructions after expiry. | The displayed 200 ms action releases automatically; repeats move again after the original action has ended. | Watch | 06:23-07:41 |
| F09 - Control-State and Conflict Management | GUI movement, Stop & release, and again. | Controls become neutral and again reports no previous movement. Deliberately reordered responses and device failures are not demonstrated. | Watch | 07:47-08:17 |
| F10 - Fall Detection and Start Recovery | GUI upward movement leaves the safe path after progress. | Fall feedback and return to the start with neutral movement; the narration explains loss of progress and cleared controls. | Watch | 08:18-09:20 |

**CLI:** 10:18-11:30 shows the terminal interface, typed directional instructions, map/state updates and fall recovery through the same local-model core. F05/F06 intentionally share the contextual reversal segment, which demonstrates both interpreting relative direction and replacing active movement.

## Independent evidence for F02, F03, and F04

- **F02 - capture/session:** GUI listening remains active across multiple spoken instructions and exposes session status and stopping.
- **F03 - transcription:** changing visible partial text appears while speech continues, independently of whether every phrase becomes a movement instruction.
- **F04 - interpretation:** GUI typed instructions reach the displayed local SemIf/Qwen backend and produce accepted movement, release and no-action feedback. The separate CLI segment shows typed gameplay without a microphone.

## Interpretation and evidence limits

The recording is a finite demonstration, not a measured recognition-accuracy or latency benchmark. Its visible delays are inference limitations, not an intentionally added game mechanic. The application's default interpretation deadline is 2000 ms; the recording does not establish a new timed performance gate. Timed movement is bounded to 100/200/400 ms: the spoken example asking for a second visibly produces a 200 ms action and should not be read as arbitrary-duration support.

The recording contains a black transition around 09:34-10:18 before the CLI. The CLI link skips it. No code walkthrough is required for the feature video, and this recording does not establish completed source/test review. Detailed device-failure testing, native Linux coverage and the updated repeat-enabled GPU timing gate remain separate limitations.

The source recording, extracted frames/audio, transcript and review helpers remain local. They are not repository submission artifacts. Only the YouTube link and timestamps are included here. Rebuild the report after any link or timing correction.

## Diagram appendices

The following pages embed the supplied UMLet SVGs as vectors without changing the diagram files. Pages use larger formats to preserve readable text at normal zoom. Source links are also indexed in section 3. These diagrams explain design and runtime contracts; they do not replace the required running-application videos.

## Appendix A - UML class diagram

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/class-diagram.svg

Say Swear - Stage 1 revised Java design
Read top to bottom within each package. Key methods are shown; concrete classes inherit interface operations.
One selected GUI/CLI shares the controller, game, controls and agent. AI receives control context only; the human chooses the route.
Typed attributes/results also express value dependencies. CompletionStage abbreviates CompletionStage<ActionDecision>.

### Presentation / MVC

«interface»
GameView
+render(snapshot: GameSnapshot)
+showMessage(message: String)
+showTranscript(text: String)
+showVoiceStatus(status: String)
+onGameEvent(event: GameEvent)

JavaFxGameView
+root(): Parent
+bindController(controller)
+requestStartVoice()
+requestStopVoice()
+render(snapshot: GameSnapshot)

CliView
-terminal: Terminal [0..1]
-status: Status [0..1]
-lineReader: DashboardReader [0..1]
-latestSnapshot: GameSnapshot
+open(quietCli: boolean): CliView
+setSessionDescription(value: String)
+showHelp()
+readCommand(): String
+setInteractive(interactive: boolean)
+render(snapshot: GameSnapshot)
+onGameEvent(event: GameEvent)
+close()
-refreshDashboard()
-updateDashboard()

Views implement GameView; GameView extends GameObserver.
CliView plans a JLine status panel and editable prompt, with a plain-terminal fallback.
Terminal, Status, Parent and CompletionStage are library types.
DashboardReader is a private CliView helper.

### Application / MVC and Observer

«interface»
GameObserver
+onGameEvent(event: GameEvent)

Main
+main(arguments: String[])
-runCli(configuration)

ApplicationController
-voiceRequested: boolean
-version, utterance: long
+start()
+submitTextCommand(text: String)
+handleTranscript(text, sourceUtterance, epoch)
+handleDecision(decision: ActionDecision)
+startVoiceControl()
+stopVoiceControl()
+handleVoiceReady(epoch)
+handleVoiceFailure(message, epoch)
+pauseControls()
+restart()
+onGameEvent(event: GameEvent)
+getSnapshot(): GameSnapshot
+close()
-submit(text, inputUtterance)
-advance(seconds: double)
-isCurrent(update: TranscriptUpdate): boolean
-failCurrent(message: String)
-invalidateControls()
-publish()

### Language interpretation / Adapter

VoiceCommandAgent
-baseContext: ControlContext
+interpret(update, context): CompletionStage
+reset()
+close()

«interface»
DecisionModel
+decide(update, baseContext): CompletionStage
+close()

OpenJevAdapter
+decide(update, baseContext): CompletionStage
+close()

DemoDecisionModel
+decide(update, baseContext): CompletionStage

loopback HTTP

«external HTTP service»
Local SemIf / Qwen3.5 4B
POST /decide
GET /health

### Deterministic controls / Command

ControlManager
-previousCommand: String
-lastAccepted: ActionDecision
-lastMovement: MovementIntent [0..1]
-lastRepeatedUtterance: long
-epoch, revision, latestVersion: long
+beginInput(version: long)
+getContext(): ControlContext
+accept(decision: ActionDecision): boolean
+validate(decision: ActionDecision): boolean
+plan(decision: ActionDecision): List<GameCommand>
+advance(deltaTime: double)
+reset()

ActionExecutor
-activeTimed: List<TimedPressCommand>
+execute(commands: List<GameCommand>)
+cancelAll()
+advance(deltaTime: double): boolean

«interface»
GameCommand
+execute(game: GameModel)
+cancel(game: GameModel)

KeyDownCommand
-direction: Direction

KeyUpCommand
-direction: Direction

ReleaseAllCommand

TimedPressCommand
-directions: Set<Direction>
-durationMs: int
-remainingMs: double
+advance(deltaTime, game): boolean

### Game model

GameModel
-heldDirections: Set<Direction>
-observers: List<GameObserver>
+update(deltaTime: double)
+evaluatePosition()
+press(direction: Direction)
+release(direction: Direction)
+releaseAll()
+resetToStart()
+restart()
+getSnapshot(): GameSnapshot
+addObserver(observer: GameObserver)
+notifyObservers(event: GameEvent)

Player
-position, velocity: Vector2
-state: PlayerState
+update(game, deltaTime)
+move(deltaTime, held)
+setState(state: PlayerState)
+respawn(position: Vector2)

Level
-safePath: List<Vector2>
-pathHalfWidth: double
-goal: Vector2
+start(): Vector2
+isSafe(position: Vector2): boolean
+isGoal(position: Vector2): boolean

GameModel is the command receiver and observer subject.
One serialized application loop owns all mutable game/control state.
Each fall cancels controls and restores Level.start(); there are no checkpoints.

### Voice input / Adapter

VoiceInputService
+start(epoch: long)
+acceptAudio(samples: float[])
+onPartial(text: String)
+onEndpoint()
-partial(session, text: String)
-endpoint(session)
+stop()
+close()

«interface»
StreamingSpeechRecognizer
+start(onPartial, onEndpoint, onError)
+acceptAudio(samples: float[])
+stop()
+close()

MicrophoneDevice
+open()
+readSamples(): float[]
+close()

SherpaOnnxAdapter
+start(onPartial, onEndpoint, onError)
+acceptAudio(samples: float[])
+stop()

### Player lifecycle / State

«interface»
PlayerState
+update(player, game, deltaTime)
+allowsMovement(): boolean
+status(): PlayerStatus

ActiveState
FallingState
RespawningState
FinishedState

### Immutable interpretation values

«record»
TranscriptUpdate
text: String
utteranceId, version, epoch: long

«record»
ControlContext
heldDirections: Set<Direction>
previousCommand: String
epoch, revision, latestVersion: long
movementAllowed: boolean
lastMovement: MovementIntent [0..1]

«record»
MovementIntent
type: ActionType
directions: Set<Direction>
durationMs: int

Memory contains resolved movement semantics.
Timer expiry and spoken stop preserve it.
Pause, fall, restart and current errors clear it.
REPEAT_LAST creates fresh commands, once per utterance.

«record»
ActionDecision
type: ActionType
directions: Set<Direction>
durationMs: int
transcriptVersion, utteranceId: long
epoch, contextRevision: long
sourceText: String
+withRequest(update, context): ActionDecision

«enumeration»
ActionType
PRESS
RELEASE
SWITCH
TIMED_PRESS
KEEP_CURRENT
RELEASE_ALL
NO_ACTION
REPEAT_LAST

«enumeration»
Direction
UP
DOWN
LEFT
RIGHT

ActionDecision carries request identity for stale-result rejection. MovementIntent deliberately contains no request identity or running timer.
Values listed by name in attributes and method signatures are shared between the packages above.

### Game values

«record»
GameSnapshot
position, velocity: Vector2
heldDirections: Set<Direction>
status: PlayerStatus
level: Level

«record»
GameEvent
type: GameEventType
snapshot: GameSnapshot

«record»
Vector2
x: double
y: double

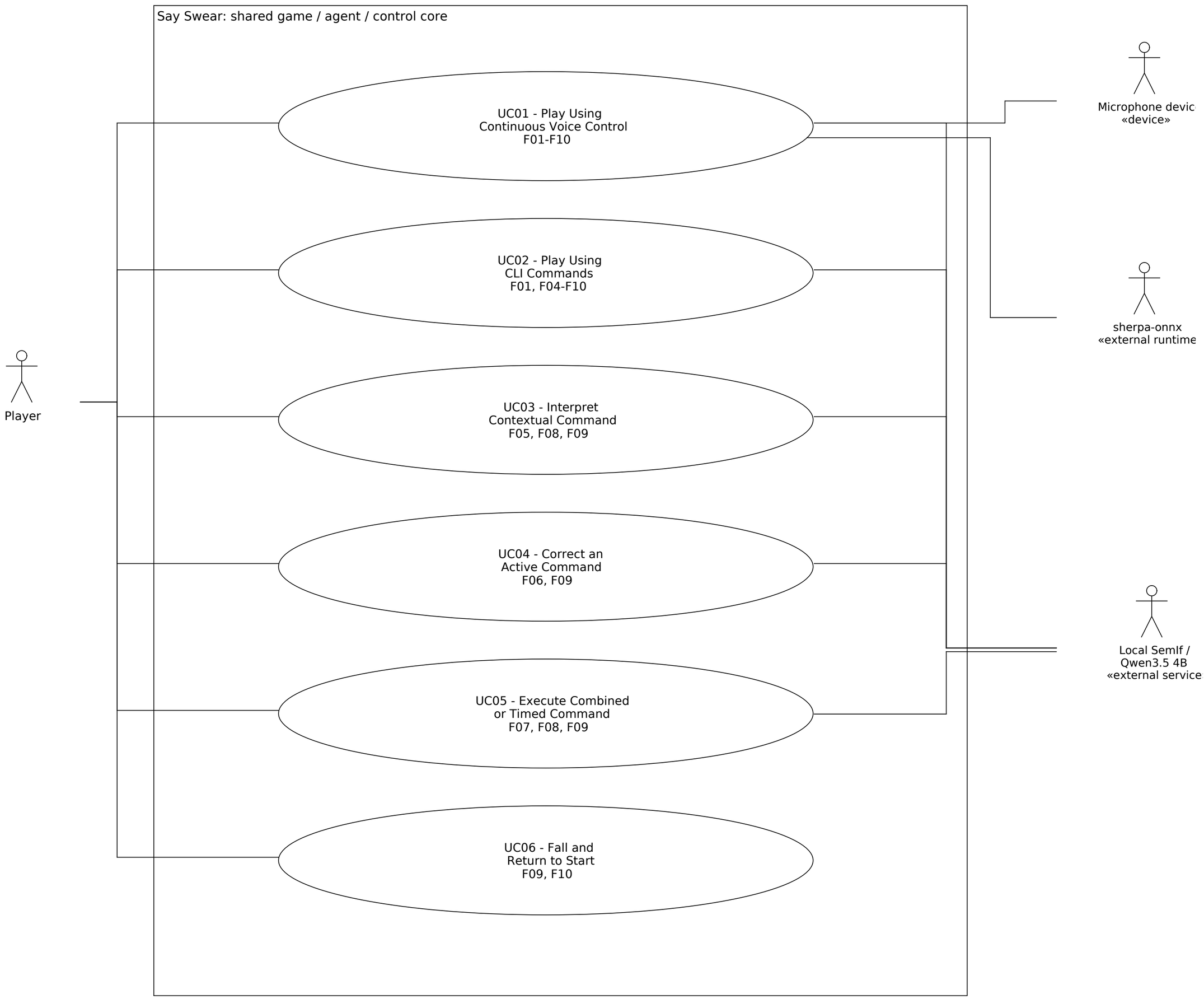
«enumeration»
PlayerStatus
ACTIVE
FALLING
RESPAWNING
FINISHED

«enumeration»
GameEventType
STATE_CHANGED
FALL_DETECTED
RESPAWNED
GOAL_REACHED

# Appendix B - Use-case diagram

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/use-case-diagram.svg

Say Swear - Stage 1 use cases
One Java application, launched with either JavaFX or a CLI frontend. UC03-UC06 occur during UC01 or UC02.
The human selects every direction; no actor or use case delegates route planning to the model.

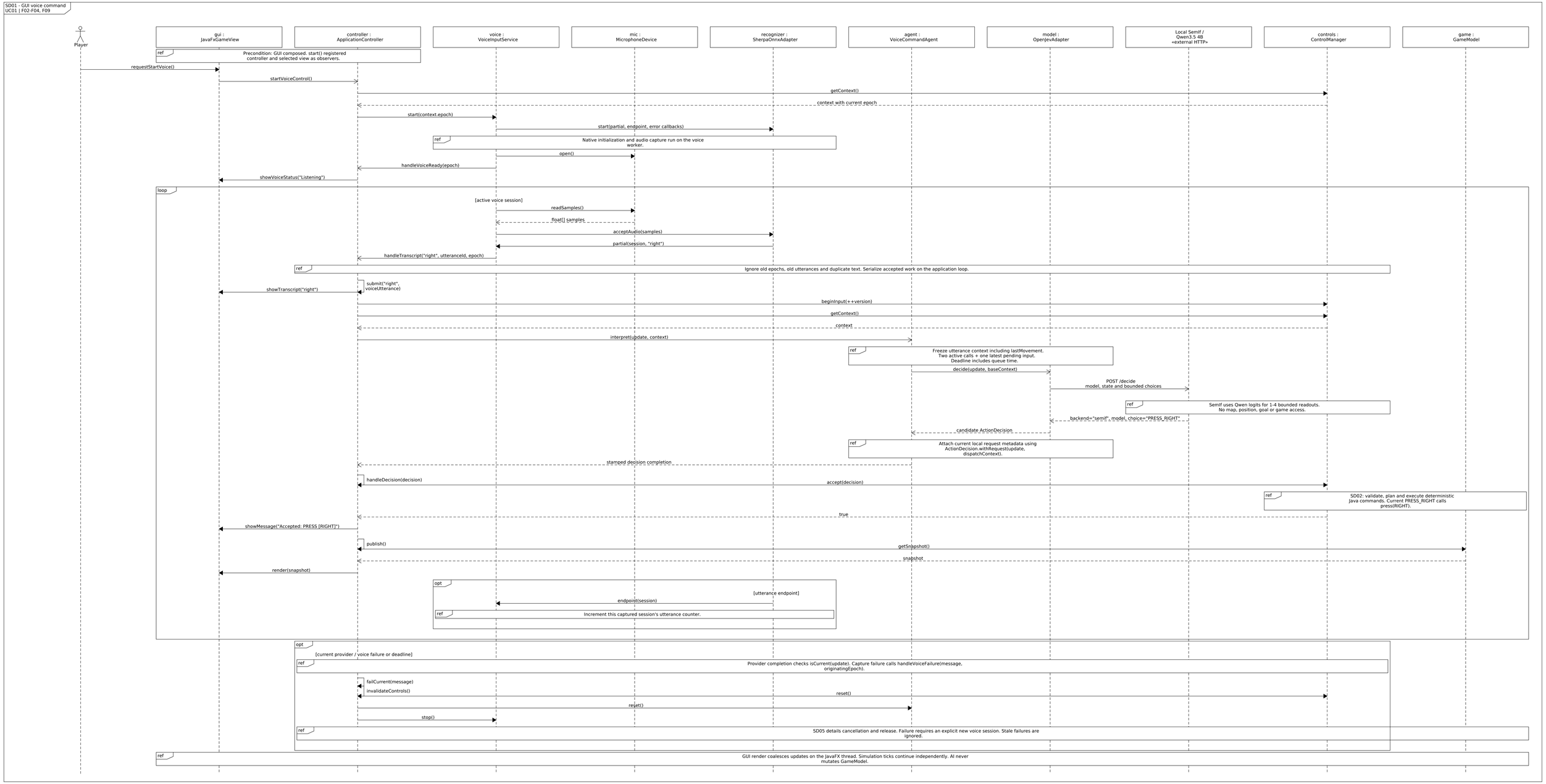



UC03 context: "other way" uses held directions;
"keep going" preserves control; "again" replays stored
MovementIntent, including a new timer after expiry.
UC04 replaces an active instruction with its newer correction.
UC05 supports diagonal or bounded-duration movement.

UC06 is triggered by leaving the safe path.
It is a system response experienced by the Player.
Java cancels commands, clears memory and returns to Level.start().
UC02 replaces microphone/transcription with typed instructions.
Actor associations show participation, not message order.

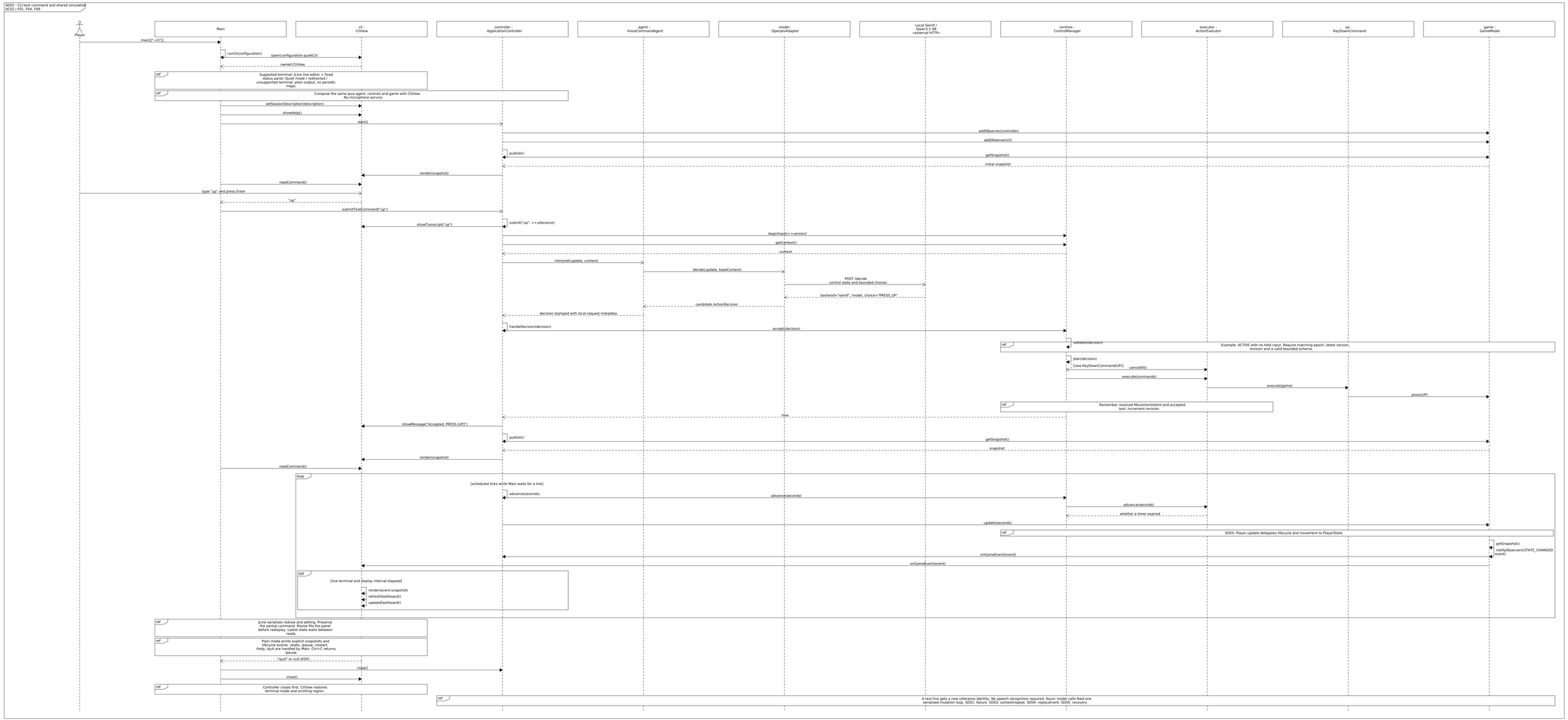
## Appendix C - SD01 GUI voice command

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/sequence-diagrams/sd01-gui-voice-command.svg



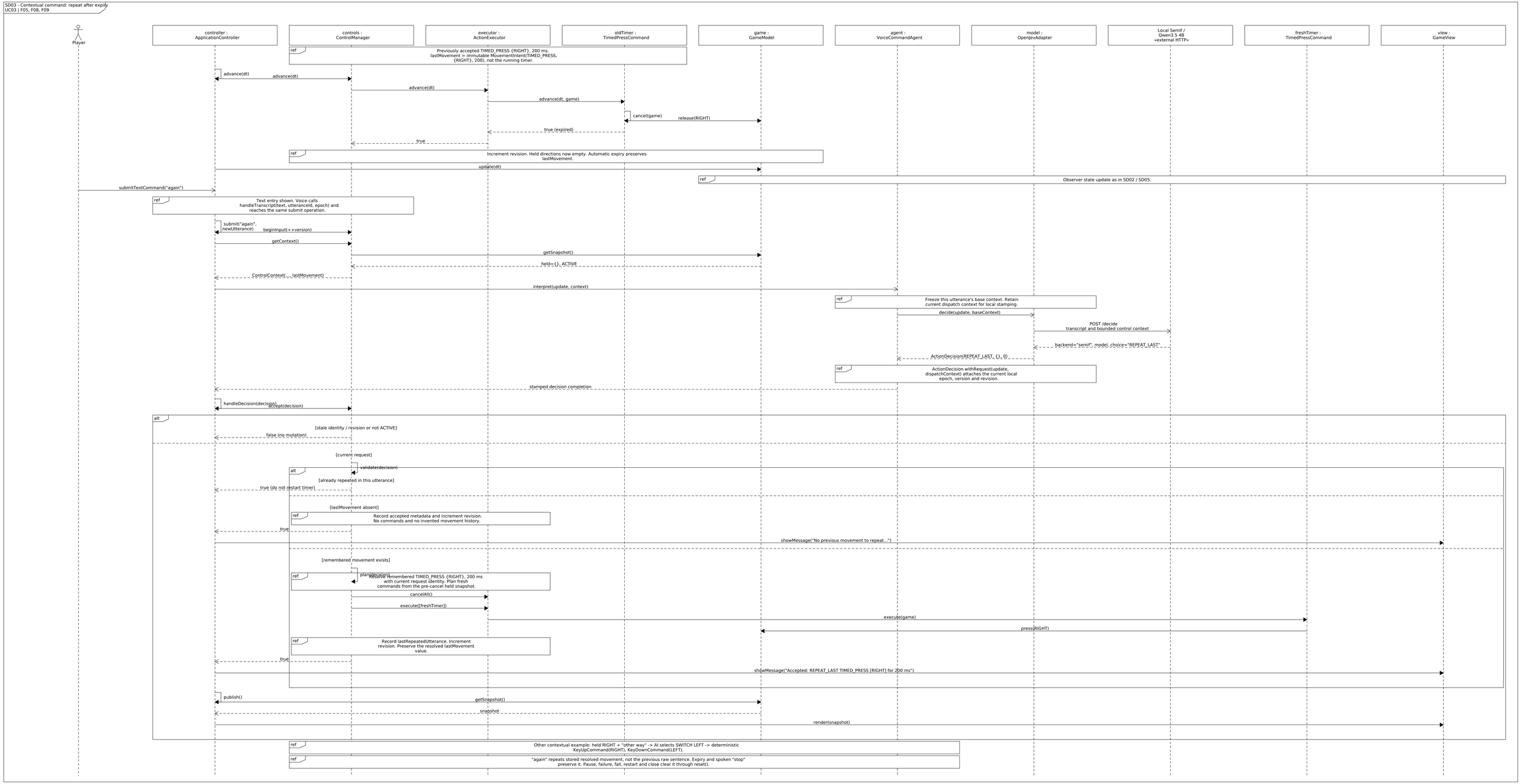
## Appendix D - SD02 CLI text command

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/sequence-diagrams/sd02-cli-text-command.svg



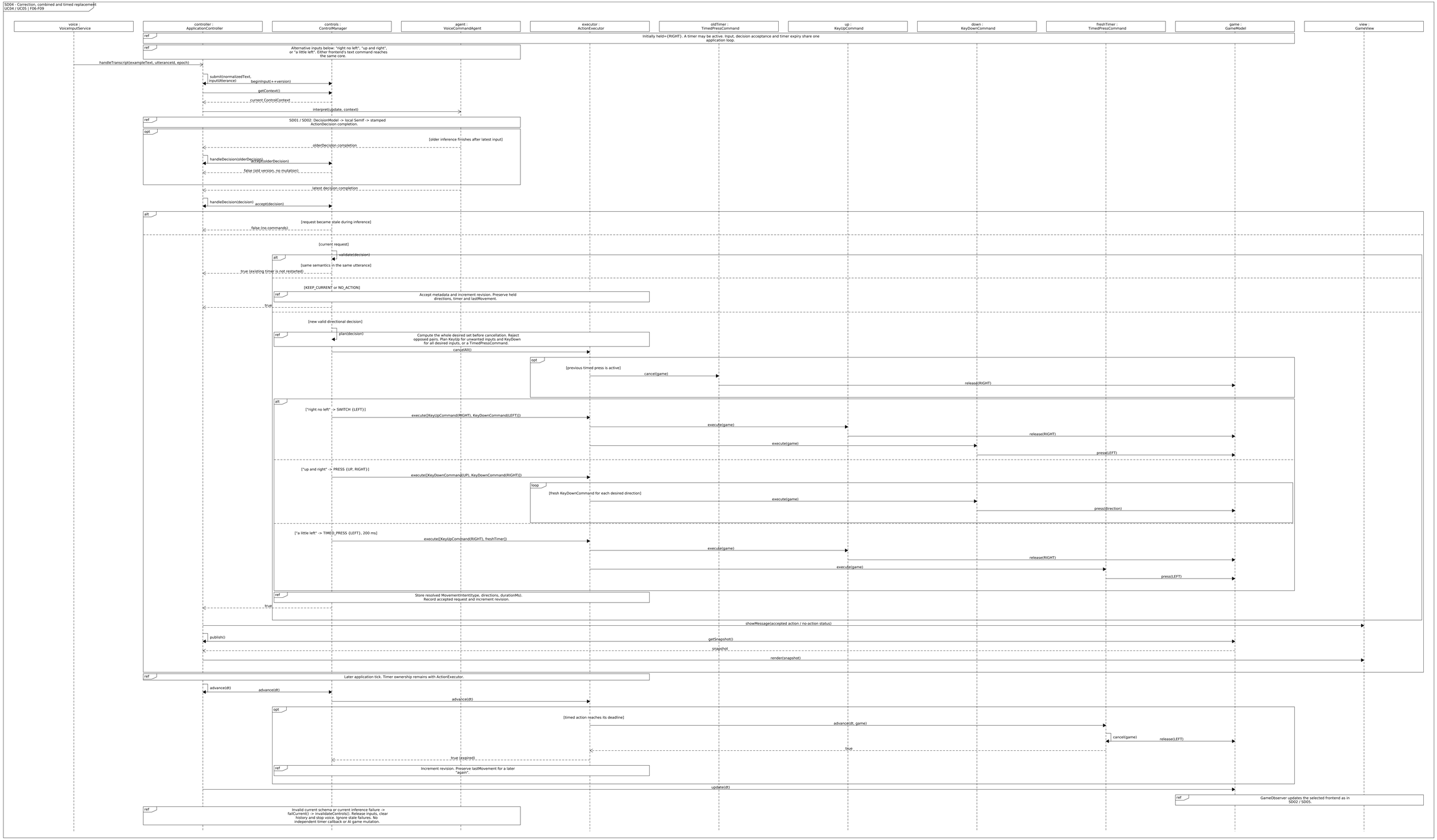
# Appendix E - SD03 contextual command and repeat

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/sequence-diagrams/sd03-contextual-command.svg



## Appendix F - SD04 correction, combined and timed commands

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/sequence-diagrams/sd04-live-correction.svg



## Appendix G - SD05 fall and start recovery

Existing UMLet SVG | large-format vector page | zoom to inspect. Source: docs/uml/sequence-diagrams/sd05-fall-recovery.svg

